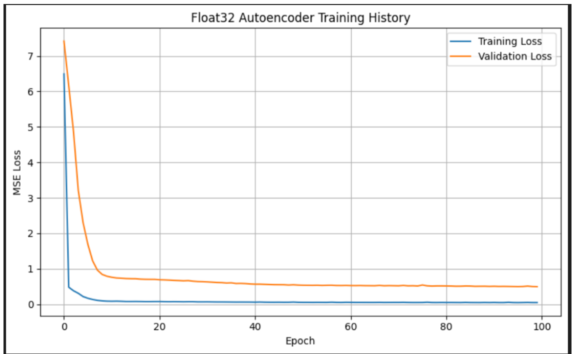
Part 1: https://studio.edgeimpulse.com/public/996777/live

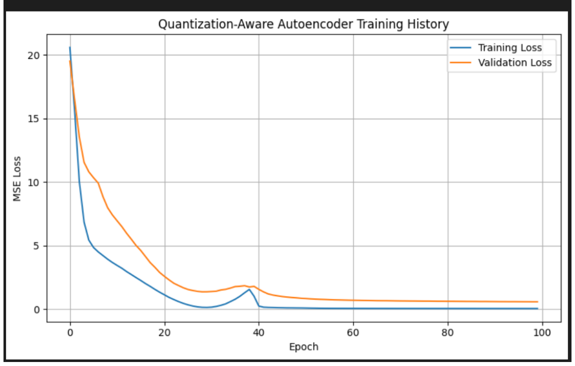
1. The raw inputs are accX, accY, accZ.
2. Type of features: Spectral Analysis Features form the accelerometer data. Number of input features: 33. Feature Names: accX RMS, accX Peak 1 Height, accZ Spectral Power 2.0 - 5.0, accY Spectral Power 2.0 - 5.0, accX Peak 1 Freq
3. Output: nominal, off
4. Type of features: spectral, Number of features: 5, accX RMS, accX Peak 1 Height, accZ Spectral Power 2.0 - 5.0, accY Spectral Power 2.0 - 5.0, accX Peak 1 Freq.
5.

```
Anomaly prediction: 16.005146
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 593 us, anomaly 6 ms, postprocessing 50 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.024487
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 594 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.024269
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 592 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
```

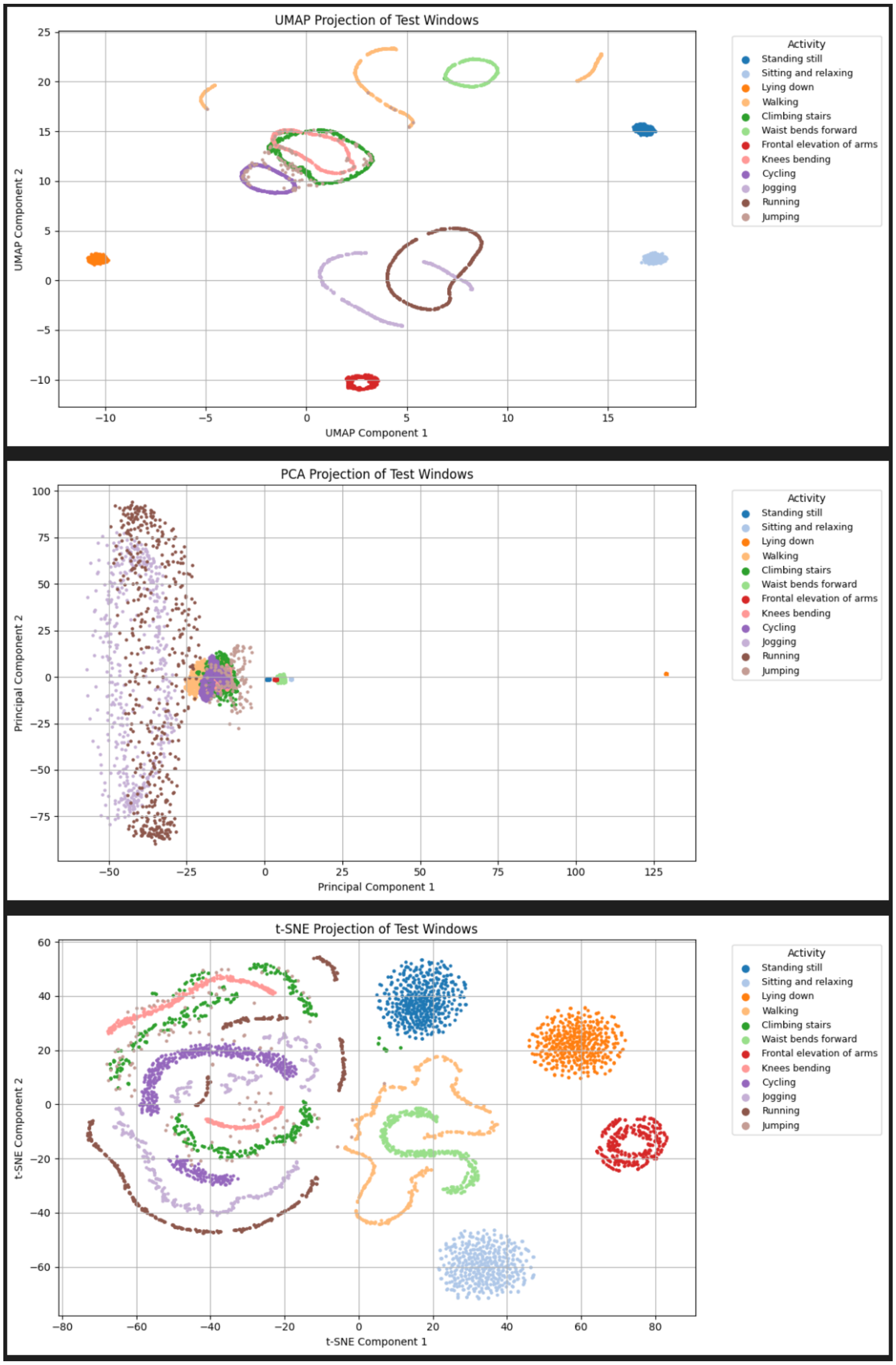
Part 2:

1. Training and loss curves:



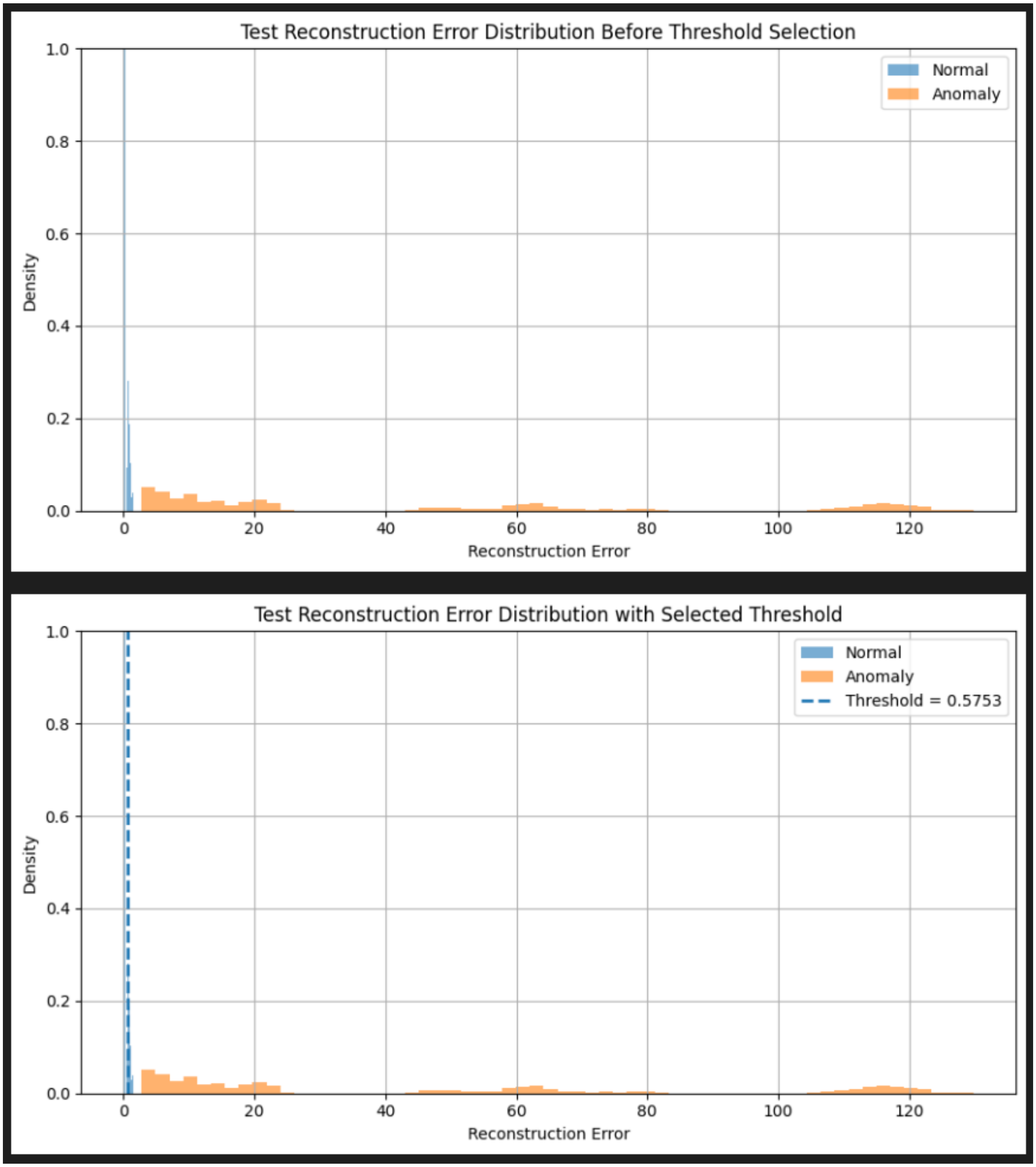


2.

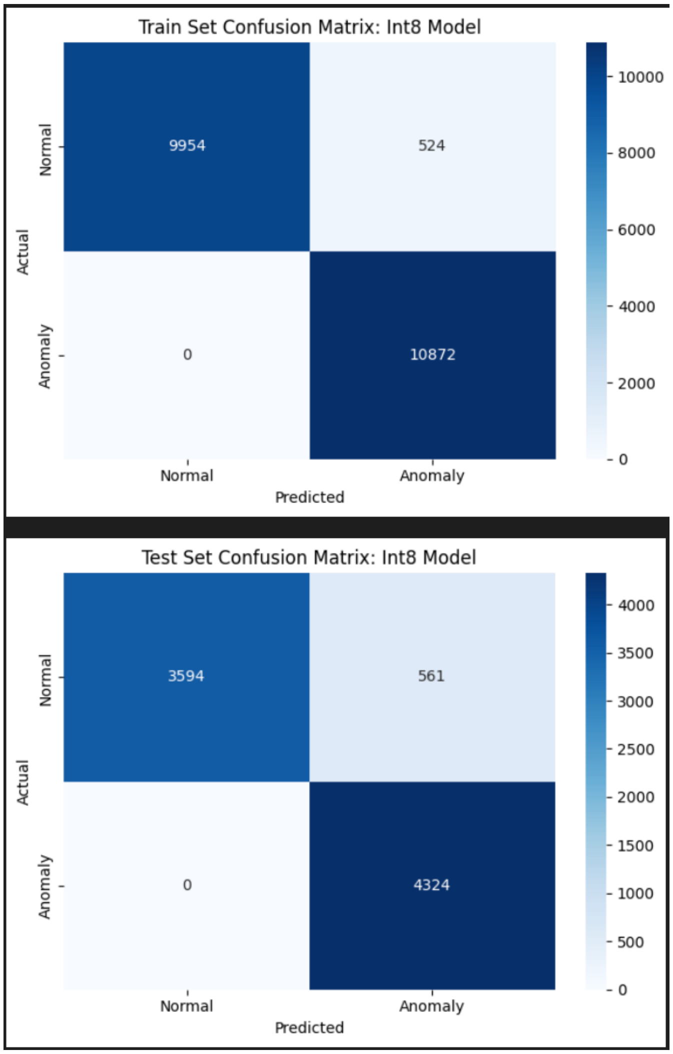


UMAP Projection of Test Windows
UMAP Component 1
UMAP Component 2
25
20
15
10
5
0
−5
−10
−10
−5
0
5
10
15
Activity
Standing still
Sitting and relaxing
Lying down
Walking
Climbing stairs
Waist bends forward
Frontal elevation of arms
Knees bending
Cycling
Jogging
Running
Jumping
PCA Projection of Test Windows
Principal Component 1
Principal Component 2
100
75
50
25
0
−25
−50
−75
−50
−25
0
25
50
75
100
125
Activity
Standing still
Sitting and relaxing
Lying down
Walking
Climbing stairs
Waist bends forward
Frontal elevation of arms
Knees bending
Cycling
Jogging
Running
Jumping
t-SNE Projection of Test Windows
t-SNE Component 1
t-SNE Component 2
60
40
20
0
−20
−40
−60
−80
−60
−40
−20
0
20
40
60
80
Activity
Standing still
Sitting and relaxing
Lying down
Walking
Climbing stairs
Waist bends forward
Frontal elevation of arms
Knees bending
Cycling
Jogging
Running
Jumping

3.


Test Reconstruction Error Distribution Before Threshold Selection
Normal
Anomaly
Density
Reconstruction Error
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
120
Test Reconstruction Error Distribution with Selected Threshold
Normal
Anomaly
Threshold = 0.5753
Density
Reconstruction Error
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
120

5.



4.

```
Reconstruction error: 0.621974
Result: Anomaly detected
Reconstruction error: 0.628025
Result: Anomaly detected
Reconstruction error: 0.569143
Result: Normal activity
Reconstruction error: 0.565034
Result: Normal activity
```

5.

Part 3:

1. We choose a threshold of 95% so that 95% of normal windows would be below this threshold.
2. It was trained on normal activity so when it is unable to reconstruct them then it triggers anomaly activity.
3. Threshold value
4. Because the Arduino could could apply different scaling, axis ordering, or window construction.
5. No retraining, limited RAM for winows sizes, fixed threshold.